

C# Programming Knowledge Assessment – Questions and Answers

Section 1: Data Types

Beginner

1. Which data type is used to store a single character?

Answer: The char data type is used to store a single Unicode character.

2. What is the default value of an uninitialized int variable in a class?

Answer: The default value is 0.

3. Which keyword is used to declare a constant variable?

Answer: The const keyword.

4. How do you convert a string to an int safely?

Answer: By using `int.TryParse()`, which prevents exceptions.

5. What is the difference between float and double?

Answer: float is 32-bit with less precision, while double is 64-bit with higher precision.

Intermediate

6. Explain the difference between Value Types and Reference Types.

Answer: Value types store data directly, whereas reference types store references to data in the heap.

7. What is Boxing and Unboxing?

Answer: Boxing converts a value type to object; unboxing extracts it back to a value type.

8. What is the purpose of the decimal type?

Answer: decimal provides high precision for financial calculations.

9. How does a struct differ from a class in memory allocation?

Answer: Structs are allocated on the stack (or inlined), classes on the heap.

10. What is a Nullable<T> type?

Answer: It allows value types to represent null values.

Advanced

11. Describe the behavior of the stackalloc keyword.

Answer: It allocates memory on the stack for performance-critical scenarios.

12. How does the dynamic type differ from object?

Answer: dynamic performs runtime type resolution instead of compile-time.

13. Explain nint and nuint.

Answer: Platform-specific native integer types.

14. What are Ref structs?

Answer: Stack-only structs like Span<T> with strict lifetime rules.

15. How does readonly struct improve performance?

Answer: Prevents defensive copying and improves immutability guarantees.

Section 2: Arrays

Beginner

1. How do you declare and initialize an array of 5 integers?

Answer: `int[] arr = new int[5];`

2. What is the index of the first element?

Answer: 0.

3. Which property gives total elements?

Answer: Length.

4. How do you change the third element?

Answer: `data[2] = value;`

5. Can an array change size?

Answer: No.

Intermediate

6. Difference between multidimensional and jagged arrays?

Answer: Multidimensional arrays have fixed dimensions; jagged arrays are arrays of arrays.

7. How do you sort an array?

Answer: `Array.Sort(array);`

8. Array.Copy vs Clone?

Answer: Copy allows partial copy; Clone copies entire array.

9. Index and Range operators?

Answer: ^1 accesses last element; .. slices a range.

10. How do you find rank?

Answer: array.Rank.

Advanced

11. How do Span<T> improve efficiency?

Answer: They avoid allocations and enable slicing.

12. foreach vs for performance?

Answer: for can be slightly faster due to bounds control.

13. What is Array Covariance?

Answer: Allowing derived type arrays to be assigned to base type arrays (can cause runtime errors).

14. Memory layout difference?

Answer: Multidimensional arrays are contiguous; jagged arrays are separate allocations.

15. Unsafe iteration using pointers?

Answer: Using fixed keyword and pointer arithmetic.

Section 3: Strings

Beginner

1. Are strings mutable?

Answer: No, strings are immutable.

2. How do you concatenate strings?

Answer: Using + operator.

3. How do you check if string contains word?

Answer: `string.Contains()`.

4. Convert to lowercase?

Answer: `ToLower()`.

5. What is String Interpolation?

Answer: Using \$ to embed expressions.

Intermediate

6. When to use `StringBuilder`?

Answer: For repeated concatenations.

7. What is String Intern Pool?

Answer: A memory optimization storing unique string literals.

8. `String.Compare` vs `==`?

Answer: `String.Compare` allows culture and case rules.

9. Purpose of @ symbol?

Answer: Verbatim strings without escape sequences.

10. Split comma-separated string?

Answer: `string.Split(',')`.

Advanced

11. `string.Create` benefits?

Answer: Reduces allocations during generation.

12. `null` vs `Empty` vs `whitespace`?

Answer: `null` is no reference, `Empty` is length 0, `whitespace` has spaces.

13. `Span<char>` benefits?

Answer: Reduces allocations while parsing.

14. How `GetHashCode` works?

Answer: Generates a hash based on string content.

15. UTF-8 string literals?

Answer: Introduced in C# 11 using `u8` suffix.

Section 4: Loops

Beginner

1. for loop syntax?

Answer: `for(int i=0;i<10;i++) { }`

2. while vs do-while?

Answer: do-while runs at least once.

3. Exit loop keyword?

Answer: break.

4. Skip iteration keyword?

Answer: continue.

5. Best loop for collections?

Answer: foreach.

Intermediate

6. Modify collection in foreach?

Answer: Throws `InvalidOperationException`.

7. Infinite loop?

Answer: `for(;;) { }`

8. Multiple variables in for?

Answer: Yes.

9. Iterator pattern?

Answer: Uses IEnumerator and MoveNext().

10. Read stream until end?

Answer: while(!stream.EndOfStream) { }

Advanced

11. What is Parallel.ForEach?

Answer: Executes iterations in parallel.

12. yield return?

Answer: Lazily returns sequence values.

13. foreach translation?

Answer: Compiled into enumerator-based while loop.

14. LINQ ForEach vs foreach?

Answer: LINQ has more overhead.

15. await foreach?

Answer: Iterates over IEnumerable asynchronously.